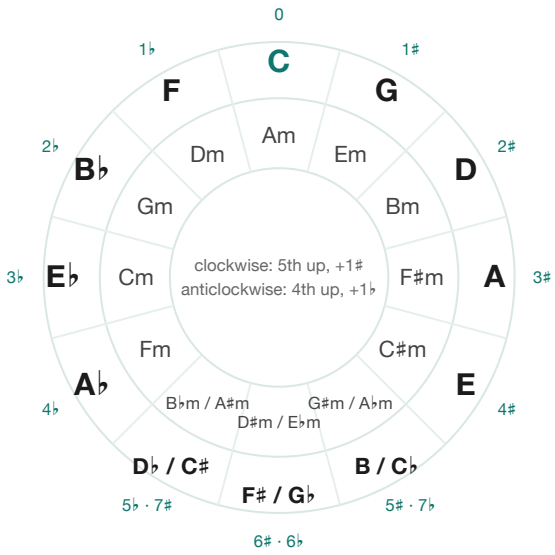
GUITAR CARDS · MEMORY PACK

Memory Essentials

Only what is worth memorising: every borrowed chord and where it comes from, the interval structure of every scale, superimposition and the circle of fifths.



1 Circle of fifths

2 Modal interchange in a major key

3 Modal interchange in a minor key

4 Interval structure of every scale

5 Superimposition — what and why

✓ Daily recall checklist

The printed companion to the Memory pack. Every chord, source mode and formula in these pages is computed by the same engine that writes the cards, so nothing is hand-typed: the borrowed-chord lists come from building the 7th chords of every parallel mode, and the scale formulas from the app's scale table. One example key (C) throughout; the numerals and formulas are what transfer to every key.

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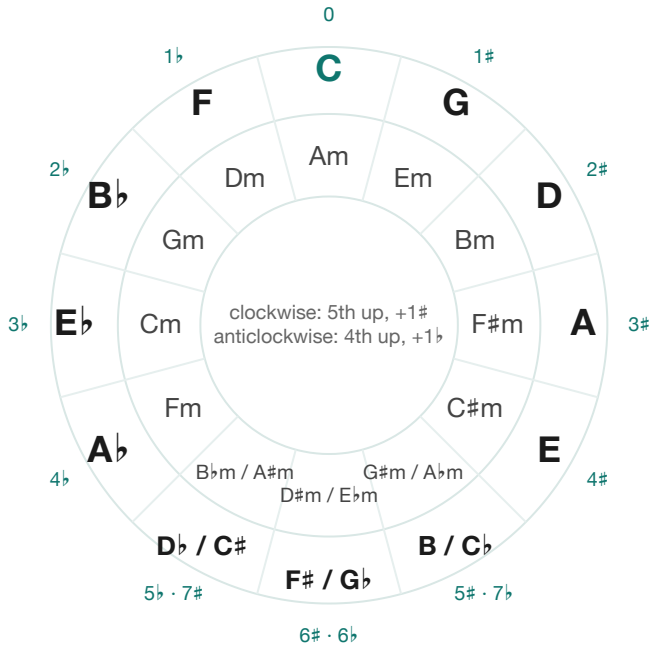
Daily recall checklist 18

- **Checklists first.** Each modal-interchange section opens with a tick-box list: name the chord, its source and its job, then tick it.
- **Roman numerals are relative to the major scale of the tonic**, as everywhere in the app: ♭VI is a minor 6th above the tonic in a major or a minor key. Lower case = minor 3rd (ivm7, iiø7, vii°7).
- **Formulas are against the major scale** of the same root: ♭3 is a minor 3rd, ♯4 an augmented 4th, 𝄫7 a diminished 7th.
- **Core** marks the chords and options that come up all the time; learn those first, then the complete set.

KEYS

1 Circle of fifths

RULE Clockwise: up a 5th, add a sharp (the new key's 7th). Anticlockwise: up a 4th, add a flat (the new key's 4th). Sharp keys: up a semitone from the last sharp. Flat keys: the second-to-last flat.



Going **clockwise** moves up a 5th and adds one sharp. The new sharp is always the new key's **7th degree**, raised so it leads to the tonic. Going **anticlockwise** moves up a 4th and adds one flat, which is always the new key's **4th degree**.

Order of sharps: **F C G D A E B** ("Father Charles Goes Down And Ends Battle"). Order of flats: the same backwards.

Two shortcuts name a key from its signature:

- **Sharp keys:** go up a semitone from the last sharp. Four sharps end on D♯, so the key is E.
- **Flat keys:** the second-to-last flat names the key. Four flats are B♭ E♭ A♭ D♭, so the key is A♭. (F, with one flat, is the one to memorise.)

Key signatures: all 15 spellings

Major key	Sig.	Accidentals, in order	Newest accidental	Rel. minor	Naming shortcut
C	—	—	—	Am	—
G	1♯	F♯	F♯ = 7th of G	Em	last sharp F♯ + semitone = G
D	2♯	F♯ C♯	C♯ = 7th of D	Bm	last sharp C♯ + semitone = D
A	3♯	F♯ C♯ G♯	G♯ = 7th of A	F♯m	last sharp G♯ + semitone = A
E	4♯	F♯ C♯ G♯ D♯	D♯ = 7th of E	C♯m	last sharp D♯ + semitone = E
B	5♯	F♯ C♯ G♯ D♯ A♯	A♯ = 7th of B	G♯m	last sharp A♯ + semitone = B
F♯	6♯	F♯ C♯ G♯ D♯ A♯ E♯	E♯ = 7th of F♯	D♯m	last sharp E♯ + semitone = F♯
C♯	7♯	F♯ C♯ G♯ D♯ A♯ E♯ B♯	B♯ = 7th of C♯	A♯m	last sharp B♯ + semitone = C♯
F	1♭	B♭	B♭ = 4th of F	Dm	one flat: memorise F
B♭	2♭	B♭ E♭	E♭ = 4th of B♭	Gm	second-to-last flat: B♭
E♭	3♭	B♭ E♭ A♭	A♭ = 4th of E♭	Cm	second-to-last flat: E♭
A♭	4♭	B♭ E♭ A♭ D♭	D♭ = 4th of A♭	Fm	second-to-last flat: A♭
D♭	5♭	B♭ E♭ A♭ D♭ G♭	G♭ = 4th of D♭	B♭m	second-to-last flat: D♭
G♭	6♭	B♭ E♭ A♭ D♭ G♭ C♭	C♭ = 4th of G♭	E♭m	second-to-last flat: G♭
C♭	7♭	B♭ E♭ A♭ D♭ G♭ C♭ F♭	F♭ = 4th of C♭	A♭m	second-to-last flat: C♭

Order of sharps: F C G D A E B · **Order of flats:** B E A D G C F (the same backwards).

Enharmonic pairs on the circle: B = C♭ (5♯ / 7♭), F♯ = G♭ (6♯ / 6♭), C♯ = D♭ (7♯ / 5♭). Sharps + flats of the pair always add to 12.

2 Modal interchange in a major key

The checklist first: tick a line when you can name the chord, its source mode and its job instantly in any key.

How the list is made. Take every parallel mode on the same tonic (Lydian, Mixolydian, Dorian, Aeolian, Phrygian, Locrian, harmonic minor, melodic minor), build the 7th chord on each degree, and drop the ones the major key already has. What is left is every borrowed chord: **27 of them**. Each appears once, with every mode it can come from. Numerals are relative to the **major** scale of the tonic, so ♭VI is always a minor 6th above the tonic.

The core 10 — know these cold

	Chord	In C	Triad form	From	When it's used
☐	**ivm7**	Fm7	**iv** Fm	Aeolian, Phrygian, Locrian, harm. minor	Minor plagal: IV–iv–I, the 6 of the key sighs down to ♭6, then to 5. The triad iv is the common form.
☐	**♭VImaj7**	A♭maj7	**♭VI** A♭ triad also in Locrian	Aeolian, Phrygian, harm. minor	Deceptive cadence V7 → ♭VImaj7 (bass 5 → ♭6); and ♭VI–♭VII–I, the triumphant "Mario" cadence.
☐	**♭VII7**	B♭7	**♭VII** B♭ triad also in Mixolydian, Dorian	Aeolian	The backdoor dominant: ivm7–♭VII7–Imaj7, a softer arrival than V7–I (♭6 → 5 instead of 7 → 1).
☐	**♭IIImaj7**	E♭maj7	**♭III** E♭ triad also in Phrygian	Dorian, Aeolian	I–♭III–IV rock and blues climb (the blues ♭3 as a chord); opens the Dameron turnaround.
☐	**iiø7**	Dm7♭5	**ii°** D°	Aeolian, harm. minor	Minor ii in a major key: iiø7–V7♭9–Imaj7, a darker, more urgent approach to I.
☐	**vm7**	Gm7	**v** Gm	Mixolydian, Dorian, Aeolian	Mixolydian v: I–v–IV–I (no leading tone, so nothing pulls); also the ii of IV (vm7–I7–IVmaj7).
☐	**♭IImaj7**	D♭maj7	**♭II** D♭	Phrygian, Locrian	Neapolitan ♭II–V7–I, or ♭IImaj7 → Imaj7 directly (the same bass as a tritone sub).
☐	**II7**	D7	**II** D	Lydian	Lydian II: I–II–IV–I as colour, not resolving to V (#4 falls back to 4). Resolving to V it is V7/V.
☐	**♭VIImaj7**	B♭maj7	**♭VII** B♭ triad also in Aeolian	Mixolydian, Dorian	I–♭VII–I and the double plagal I–♭VII–IV–I; the maj7 form is the Mixolydian/Dorian colour.
☐	**im7**	Cm7	**i** Cm triad also in harm. minor, mel. minor	Dorian, Aeolian, Phrygian	The parallel switch: Imaj7 → im7, a whole section moves to the minor side of the same tonic.

Variant to know: the jazz form of iv is **ivm6** (Fm6 in C, from Aeolian, harm. minor); its notes double as a rootless ♭VII9, the backdoor dominant.

The rest — the complete set

	Chord	In C	Triad form	From	When it's used
☐	**vii°7**	B°7	same triad as the key (vii°); only the 7th is borrowed	harm. minor	Leading-tone diminished into I (vii°7–I); also a common-tone passing chord.
☐	**I7**	C7	same triad as the key (I); only the 7th is borrowed	Mixolydian	Same chord as V7/IV: I7–IVmaj7; and the blues tonic.
☐	**IV7**	F7	same triad as the key (IV); only the 7th is borrowed	Dorian, mel. minor	The blues IV7 (I7–IV7), and IV7–I as a bluesy plagal cadence.
☐	**♭VI7**	A♭7	**♭VI** A♭ triad also in Aeolian, Phrygian, harm. minor	Locrian	Tritone sub of II7 (V7/V): ♭VI7–V7; also ♭VI7–I as a backdoor variant.
☐	**♭viim7**	B♭m7	**♭vii** B♭m	Phrygian, Locrian	ii of ♭VI: ♭viim7–♭III7–♭VImaj7, a common modulating bridge move.
☐	**♭III7**	E♭7	**♭III** E♭ triad also in Dorian, Aeolian	Phrygian	Tritone sub of VI7 (V7/ii): ♭III7–iim7; and V7 of ♭VI.
☐	**viø7**	Am7♭5	**vi°** A°	Dorian, mel. minor	Minor ii–V into V: viø7–II7–V7. Same notes as im6 with the 6 in the bass.
☐	**i(maj7)**	Cm(maj7)	**i** Cm triad also in Dorian, Aeolian, Phrygian	harm. minor, mel. minor	Minor line cliché on the parallel-minor tonic: im–i(maj7)–im7–im6.
☐	**♭iiim7**	E♭m7	**♭iii** E♭m	Locrian	Chromatic ii–Vs falling: iiim7–♭iiim7–iim7; ii of the ♭VI7.
☐	**vø7**	Gm7♭5	**v°** G°	Phrygian	ii of a minor iv: vø7–I7♭9–ivm7.
☐	**♯ivø7**	F♯m7♭5	**♯iv°** F♯°	Lydian	Rising bass IV–♯ivø7–I/5 (4 → ♯4 → 5); ii of iii: ♯ivø7–VII7–iiim7.
☐	**iiiø7**	Em7♭5	**iii°** E°	Mixolydian	ii of a minor ii: iiiø7–VI7♭9–iim7.
☐	**iø7**	Cm7♭5	**i°** C°	Locrian	ii of ♭VII: iø7–IV7–♭VIImaj7. Rare as a tonic.
☐	**viim7**	Bm7	**vii** Bm	Lydian	Rare: Lydian colour; a brighter ii into vi (viim7–III7–vim7).
☐	**Vmaj7**	Gmaj7	same triad as the key (V); only the 7th is borrowed	Lydian	Rare: Lydian vamps (Imaj7–Vmaj7); no ♭7, so no dominant pull.
☐	**♭Vmaj7**	G♭maj7	**♭V** G♭	Locrian	Rare: a distant, dark colour a tritone from the tonic (film, modern writing).
☐	**♭IIImaj7♯5**	E♭maj7♯5	**♭III+** E♭+	harm. minor, mel. minor	Rare: really a rootless i(maj9); a colour voicing of the minor tonic.

What each parallel mode lends

Each row: the notes that differ from major, and every chord that changes because of them. Read it as “from Aeolian I get im7, iiø7, ♭IIImaj7, ivm7, vm7, ♭VImaj7, ♭VII7”.

Parallel mode	vs major	Chords it lends (bold = core)
Lydian	♯4	**II7** D7, ♯ivø7 F♯m7♭5, Vmaj7 Gmaj7, viim7 Bm7
Mixolydian	♭7	I7 C7, iiiø7 Em7♭5, **vm7** Gm7, **♭VIImaj7** B♭maj7
Dorian	♭3 ♭7	**im7** Cm7, **♭IIImaj7** E♭maj7, IV7 F7, **vm7** Gm7, viø7 Am7♭5, **♭VIImaj7** B♭maj7
Aeolian	♭3 ♭6 ♭7	**im7** Cm7, **iiø7** Dm7♭5, **♭IIImaj7** E♭maj7, **ivm7** Fm7, **vm7** Gm7, **♭VImaj7** A♭maj7, **♭VII7** B♭7
Phrygian	♭2 ♭3 ♭6 ♭7	**im7** Cm7, **♭IImaj7** D♭maj7, ♭III7 E♭7, **ivm7** Fm7, vø7 Gm7♭5, **♭VImaj7** A♭maj7, ♭viim7 B♭m7
Locrian	♭2 ♭3 ♭5 ♭6 ♭7	iø7 Cm7♭5, **♭IImaj7** D♭maj7, ♭iiim7 E♭m7, **ivm7** Fm7, ♭Vmaj7 G♭maj7, ♭VI7 A♭7, ♭viim7 B♭m7
harmonic minor	♭3 ♭6	i(maj7) Cm(maj7), **iiø7** Dm7♭5, ♭IIImaj7♯5 E♭maj7♯5, **ivm7** Fm7, **♭VImaj7** A♭maj7, vii°7 B°7
melodic minor	♭3	i(maj7) Cm(maj7), ♭IIImaj7♯5 E♭maj7♯5, IV7 F7, viø7 Am7♭5

> **Finding a source fast:** a borrowed chord needs its altered notes. ♭VI contains ♭6 (and ♭3), so it comes from a mode with ♭6: Aeolian, Phrygian or harmonic minor. II7 contains ♯4, which only Lydian has. The darker the mode (brightness ladder), the flatter the chords it lends.

In context: the stock moves

The moves from the Modal-interchange core set, one line each. Learn the numerals, the source and the one voice that moves.

Move	Chart	From	The voice that moves	Where you hear it
V7 → ♭VImaj7 → V7 → I	**iim7 V7 | ♭VImaj7 | V7 | Imaj7**	Aeolian	5 → ♭6 → 5 in the bass; the tonic note (the 3rd of ♭VI) is held on top, so a melody can still “land” on 1.	Ballad endings, film cues, and any time a song wants one more phrase before it ends.
IV → iv → I (minor plagal)	**I | IV | iv | I**	Aeolian / harmonic minor	6 → ♭6 → 5: one voice walks down chromatically.	The bittersweet “amen”: pop ballads, Beatles-era songwriting, gospel endings.
IVmaj7 → iv6 → Imaj7 (jazz)	**Imaj7 | IVmaj7 | ivm6 | Imaj7**	Aeolian / harmonic minor	6 → ♭6 → 5 again, plus iv6 shares three notes with ♭VII9.	Standards and ballads, often in bar 5–6 of a 32-bar A section.
I → ♭VII → IV → I (double plagal)	**I | ♭VII | IV | I**	Mixolydian	♭7 → 6 → 5 in the top voice, or the root motion falling in 4ths.	Rock anthems and classic rock riffs; the sound of ♭VII instead of V.
♭VI → ♭VII → I (the “Mario” cadence)	**I | ♭VI | ♭VII | I**	Aeolian	The bass walks ♭6 → ♭7 → 1; on top, 1 → 2 → 3 or 3 → 4 → 5.	Video game victory themes, rock and film climaxes.
Imaj7 → ♭VImaj7 → ♭VII7 → Imaj7	**Imaj7 | ♭VImaj7 | ♭VII7 | Imaj7**	Aeolian	The 3rd of ♭VII7 (2 of the key) and its 7th (♭6) resolve to 3 and 5 of I.	Endings and turnarounds in standards; also a common last-bars reharmonisation.
iv7 → ♭VII7 → I (backdoor ii–V)	**ivm7 ♭VII7 | Imaj7 | %**	Aeolian	♭6 → 5 (instead of the leading tone 7 → 1).	Jazz standards, especially the turn into a bridge or the final cadence.
iiø7 → V7♭9 → Imaj7 (minor ii in a major key)	**iiø7 V7♭9 | Imaj7 | %**	Aeolian / harmonic minor	The ♭5 of iiø (♭6 of the key) stays as the ♭9 of V, then falls to 5.	Standards and bebop heads; a stock reharmonisation of any ii–V–I.

Move	Chart	From	The voice that moves	Where you hear it
I → ♭III → IV → I	**I** | **♭III** | **IV** | **I**	Aeolian	The bass climbs 1 → ♭3 → 4; the ♭3 against the major tonic is the blues sound.	Rock and blues-rock riffs, soul, garage rock.
I → ♭II → V7 → I (Neapolitan)	**I** | **♭II** | **V7** | **I**	Phrygian	♭2 → 7 → 1: the ♭2 falls a diminished 3rd to the leading tone, then resolves.	Classical cadences, film music, flamenco and metal.
I → v → IV → I (Mixolydian)	**I** | **v** | **IV** | **I**	Mixolydian	♭7 (the 3rd of v) instead of 7: nothing “needs” to resolve.	Folk, Celtic, 60s and 70s rock, modal pop.
I → II → IV → I (Lydian II)	**I** | **II** | **IV** | **I**	Lydian	♯4 (the 3rd of II) falls back to the natural 4 of IV.	Beatles-era pop, film themes, dreamy singer-songwriter writing.
Imaj7 → ♭IIImaj7 → ♭VImaj7 → ♭IImaj7 (Dameron)	**Imaj7** **♭IIImaj7** | **♭VImaj7** **♭IImaj7** | **Imaj7**	Aeolian + Phrygian	The bass falls in 5ths from ♭III and slides a half step into I.	Bebop and modern jazz turnarounds (the ending of Dameron’s Lady Bird).
Imaj7 → i7 (switch to the parallel minor)	**Imaj7** | **%** | **im7** | **%**	Dorian / Aeolian	3 → ♭3 (and 7 → ♭7) over a static root.	Standards that move from a major A section to a parallel-minor B section; film music mood changes.

3 Modal interchange in a minor key

Same method, with natural minor as home. Numerals are still relative to the major scale of the tonic: the natural-minor chords read im7 iiø7 ♭IIImaj7 ivm7 vm7 ♭VImaj7 ♭VII7.

Home scale: natural minor (Aeolian). Its own chords: im7 Cm7 · iiø7 Dm7♭5 · ♭IIImaj7 E♭maj7 · ivm7 Fm7 · vm7 Gm7 · ♭VImaj7 A♭maj7 · ♭VII7 B♭7. Everything below is borrowed from the other parallel modes on the same tonic: the parallel major (Ionian), Dorian, Phrygian, Lydian, Mixolydian, Locrian, harmonic minor and melodic minor. **27 chords**, each listed once with all its sources.

The core 10 — know these cold

	Chord	In C minor	Triad form	From	When it's used
☐	**V7**	G7	**V** G triad also in Lydian	Ionian, harm. minor, mel. minor	The real dominant: V7–im, usually V7♭9; minor ii–V–i = iiø7–V7–im.
☐	**vii°7**	B°7	**vii°** B° triad also in Ionian, mel. minor	harm. minor	Leading-tone diminished: vii°7–im; it is V7♭9 without its root.
☐	**Imaj7**	Cmaj7	**I** C triad also in Mixolydian	Ionian, Lydian	Picardy third: end a minor piece on a major I (triad or maj7); the switch to the parallel major.
☐	**IV7**	F7	**IV** F triad also in Ionian, Mixolydian	Dorian, mel. minor	Dorian vamp im7–IV7: the ♮6 (3rd of IV) brightens the minor. Funk, soul-jazz, modal tunes.
☐	**iim7**	Dm7	**ii** Dm	Ionian, Mixolydian, Dorian, mel. minor	Dorian / melodic-minor ii: iim7–V7–im, softer than iiø7; iim7–IV7 vamps.
☐	**♭IImaj7**	D♭maj7	**♭II** D♭	Phrygian, Locrian	Neapolitan ♭II–V7–im, or ♭II–im directly (Phrygian, flamenco, metal).
☐	**i(maj7)**	Cm(maj7)	same triad as the key (i); only the 7th is borrowed	harm. minor, mel. minor	The jazz minor tonic and the line cliché im–i(maj7)–im7–im6.
☐	**viø7**	Am7♭5	**vi°** A°	Dorian, mel. minor	Same notes as im6 with the ♮6 in the bass: line clichés and Dorian minor tunes.
☐	**I7**	C7	**I** C triad also in Ionian, Lydian	Mixolydian	V7/iv: I7–ivm7, the major-3rd turn into iv.
☐	**♭VI7**	A♭7	same triad as the key (♭VI); only the 7th is borrowed	Locrian	Augmented-6th sound: ♭VI7–V7, the tritone sub of II7 (V7/V).

Variant to know: the jazz minor tonic **im6** (Cm6 in C minor, from Dorian, mel. minor).

The rest — the complete set

	Chord	In C minor	Triad form	From	When it's used
☐	**II7**	D7	**II** D	Lydian	V7/V: II7–V7–im; from Lydian it is pure colour.
☐	**IVmaj7**	Fmaj7	**IV** F triad also in Dorian, mel. minor	Ionian, Mixolydian	The major IV as a maj7: brighter, rarer than IV7.
☐	**♭VIImaj7**	B♭maj7	same triad as the key (♭VII); only the 7th is borrowed	Mixolydian, Dorian	♭VII as a maj7 instead of the natural ♭VII7: softer, for modal minor vamps.
☐	**viiø7**	Bm7♭5	**vii°** B° triad also in harm. minor	Ionian, mel. minor	A rootless V9: passing chord into im.
☐	**♭III7**	E♭7	same triad as the key (♭III); only the 7th is borrowed	Phrygian	V7 of ♭VI: ♭III7–♭VImaj7; tritone sub of VI7.
☐	**♭viim7**	B♭m7	**♭vii** B♭m	Phrygian, Locrian	ii of ♭VI: ♭viim7–♭III7–♭VImaj7.
☐	**vø7**	Gm7♭5	**v°** G°	Phrygian	ii of iv: vø7–I7♭9–ivm7.
☐	**♯ivø7**	F♯m7♭5	**♯iv°** F♯°	Lydian	Rising bass iv–♯ivø7–V (4 → ♯4 → 5).
☐	**iiim7**	Em7	**iii** Em	Ionian, Lydian	Parallel-major colour: appears when the music turns to the major side.
☐	**vim7**	Am7	**vi** Am	Ionian, Lydian, Mixolydian	Parallel-major vi; with the ♮6 in the bass it leans Dorian.
☐	**iiiø7**	Em7♭5	**iii°** E°	Mixolydian	ii of ii: iiiø7–VI7–iim7.
☐	**iø7**	Cm7♭5	**i°** C°	Locrian	The darkest tonic (Locrian); ii of ♭VII: iø7–IV7–♭VII.
☐	**♭iiim7**	E♭m7	**♭iii** E♭m	Locrian	ii of ♭II: ♭iiim7–♭VI7–♭IImaj7.
☐	**Vmaj7**	Gmaj7	**V** G triad also in Ionian, harm. minor, mel. minor	Lydian	Rare: Lydian colour; no ♭7, so no dominant pull.
☐	**viim7**	Bm7	**vii** Bm	Lydian	Rare: Lydian colour.
☐	**♭Vmaj7**	G♭maj7	**♭V** G♭	Locrian	Rare: tritone colour (film, metal).
☐	**♭IIImaj7♯5**	E♭maj7♯5	**♭III+** E♭+	harm. minor, mel. minor	Rare: the harmonic/melodic-minor ♭III, really a rootless i(maj9).

What each parallel mode lends

Against natural minor. Harmonic and melodic minor are the classical and jazz ways of getting a real V7 (♮7); Dorian and melodic minor give the major IV (♮6); the parallel major gives the Picardy I (♮3).

Parallel mode	vs natural minor	Chords it lends (bold = core)
Ionian (major)	♮3 ♮6 ♮7	**Imaj7** Cmaj7, **iim7** Dm7, iiim7 Em7, IVmaj7 Fmaj7, **V7** G7, vim7 Am7, viiø7 Bm7♭5
Lydian	♮3 #4 ♮6 ♮7	**Imaj7** Cmaj7, II7 D7, iiim7 Em7, #ivø7 F♯m7♭5, Vmaj7 Gmaj7, vim7 Am7, viim7 Bm7
Mixolydian	♮3 ♮6	**I7** C7, **iim7** Dm7, iiiø7 Em7♭5, IVmaj7 Fmaj7, vim7 Am7, ♭VIImaj7 B♭maj7
Dorian	♮6	**iim7** Dm7, **IV7** F7, **viø7** Am7♭5, ♭VIImaj7 B♭maj7
Phrygian	♭2	**♭IImaj7** D♭maj7, ♭III7 E♭7, vø7 Gm7♭5, ♭viim7 B♭m7
Locrian	♭2 ♭5	iø7 Cm7♭5, **♭IImaj7** D♭maj7, ♭iiim7 E♭m7, ♭Vmaj7 G♭maj7, **♭VI7** A♭7, ♭viim7 B♭m7
harmonic minor	♮7	**i(maj7)** Cm(maj7), ♭IIImaj7♯5 E♭maj7♯5, **V7** G7, **vii°7** B°7
melodic minor	♮6 ♮7	**i(maj7)** Cm(maj7), **iim7** Dm7, ♭IIImaj7♯5 E♭maj7♯5, **IV7** F7, **V7** G7, **viø7** Am7♭5, viiø7 Bm7♭5

In context

Move	Chart	From	The voice that moves	Where you hear it
i7 → IV7 (Dorian vamp, minor key)	**im7** | **IV7** | **im7** | **IV7**	Dorian	The natural 6 (the 3rd of IV) against the minor tonic: 6 → 5 or ♭7.	Funk, soul-jazz, Latin and modal jazz vamps.
i → iv → V7 → I (Picardy third)	**i** | **iv** | **V7** | **I**	Ionian (borrowed into minor)	♭3 → 3 at the last moment.	Baroque and classical endings, hymns, folk songs in minor.

4 Interval structure of every scale

Every formula is written against the major scale of the same root: 1 2 3 4 5 6 7 is major, ♭3 means a minor 3rd above the root, and so on. The 7th chord of each mode is its 1st, 3rd, 5th and 7th notes, so it is read straight off the formula.

Major modes

#	Mode	Formula (against major)	7th chord	Characteristic	Use	Brightness
1	**Ionian**	1 2 3 4 5 6 7	1 3 5 7 = **maj7**	♮4, ♮7	I chord: maj7, 6, 6/9 (the 4th is the avoid note)	2
2	**Dorian**	1 2 ♭3 4 5 6 ♭7	1 ♭3 5 ♭7 = **m7**	♮6	ii chord; minor vamps (m7, m6, m9)	4
3	**Phrygian**	1 ♭2 ♭3 4 5 ♭6 ♭7	1 ♭3 5 ♭7 = **m7**	♭2	iii chord; Spanish / metal minor	6
4	**Lydian**	1 2 3 #4 5 6 7	1 3 5 7 = **maj7**	#4	IV chord; the bright maj7#11 tonic	1
5	**Mixolydian**	1 2 3 4 5 6 ♭7	1 3 5 ♭7 = **7**	♭7 with ♮3	V chord (unaltered 7, 9, 13, sus); blues	3
6	**Aeolian**	1 2 ♭3 4 5 ♭6 ♭7	1 ♭3 5 ♭7 = **m7**	♭6	vi chord; the natural minor key	5
7	**Locrian**	1 ♭2 ♭3 4 ♭5 ♭6 ♭7	1 ♭3 ♭5 ♭7 = **m7♭5**	♭5 (and ♭2)	viiø7; m7♭5 (Locrian ♮2 is usually preferred)	7

Brightness ladder: Lydian #4 → Ionian → Mixolydian ♭7 → Dorian ♭3 → Aeolian ♭6 → Phrygian ♭2 → Locrian ♭5. Each step darkens one more note.

Melodic minor modes

#	Mode	Formula (against major)	7th chord	Characteristic	Use
1	**Melodic minor**	1 2 ♭3 4 5 6 7	1 ♭3 5 7 = **m(maj7)**	6, 9	Tonic minor (m6/9)
2	**Dorian ♭2**	1 ♭2 ♭3 4 5 6 ♭7	1 ♭3 5 ♭7 = **m7**	♭9	7sus♭9 sound
3	**Lydian augmented**	1 2 3 #4 #5 6 7	1 3 #5 7 = **maj7#5**	#11	Floating maj7#5
4	**Lydian dominant**	1 2 3 #4 5 6 ♭7	1 3 5 ♭7 = **7**	#11	IV7, ♭VII7, tritone subs
5	**Mixolydian ♭6**	1 2 3 4 5 ♭6 ♭7	1 3 5 ♭7 = **7**	♭13	Gentle V7 in minor
6	**Locrian ♮2**	1 2 ♭3 4 ♭5 ♭6 ♭7	1 ♭3 ♭5 ♭7 = **m7♭5**	♮9	iiø7 in minor (standard)
7	**Altered**	1 ♭2 ♭3 ♭4 ♭5 ♭6 ♭7 as a dominant: 1 ♭9 #9 3 #11 ♭13 ♭7	1 ♭3 ♭5 ♭7 = **m7♭5** read as 7alt: ♭4 = 3	♭9, #9, #11, ♭13	Tense V7 into major or minor

Melodic minor = major with ♭3. Each mode is a major mode with one note changed: Lydian dominant = Mixolydian #4, Altered = Locrian ♭4, Locrian ♮2 = Locrian with ♮2.

The chords of melodic minor

Degree	Chord	Mode	7th chord as intervals	Symbol	In C melodic minor
1	**i(maj7)**	Melodic minor	1 ♭3 5 7	**m(maj7)** minor-major 7	Cm(maj7)
2	**iim7**	Dorian ♭2	1 ♭3 5 ♭7	**m7** minor 7	Dm7
♭3	**♭IIImaj7♯5**	Lydian augmented	1 3 ♯5 7	**maj7♯5** major 7 ♯5	E♭maj7♯5
4	**IV7**	Lydian dominant	1 3 5 ♭7	**7** dominant 7	F7
5	**V7**	Mixolydian ♭6	1 3 5 ♭7	**7** dominant 7	G7
6	**viø7**	Locrian ♮2	1 ♭3 ♭5 ♭7	**m7♭5** half-diminished	Am7♭5
7	**viiø7**	Altered	1 ♭3 ♭5 ♭7	**m7♭5** / 7alt half-diminished	Bm7♭5

Two pairs repeat: IV7 and V7 are both 1 3 5 ♭7 (Lydian dominant and Mixolydian ♭6), and viø7 and viiø7 are both 1 ♭3 ♭5 ♭7 (Locrian ♮2 and Altered, which is played as 7alt).

Harmonic minor modes

#	Mode	Formula (against major)	7th chord	Characteristic	Use
1	**Harmonic minor**	1 2 ♭3 4 5 ♭6 7	1 ♭3 5 7 = **m(maj7)**	♮7	Dark tonic minor
2	**Locrian ♮6**	1 ♭2 ♭3 4 ♭5 6 ♭7	1 ♭3 ♭5 ♭7 = **m7♭5**	♮13	iiø7 in minor
3	**Ionian ♯5**	1 2 3 4 ♯5 6 7	1 3 ♯5 7 = **maj7♯5**	♯5	Augmented colour
4	**Dorian ♯4**	1 2 ♭3 ♯4 5 6 ♭7	1 ♭3 5 ♭7 = **m7**	♯11	Exotic iv
5	**Phrygian dominant**	1 ♭2 3 4 5 ♭6 ♭7 as a dominant: 1 ♭9 3 4 5 ♭13 ♭7	1 3 5 ♭7 = **7**	♭9, ♭13	V7♭9 into minor
6	**Lydian ♯2**	1 ♯2 3 ♯4 5 6 7	1 3 5 7 = **maj7**	♯9, ♯11	Exotic ♭VI
7	**Ultralocrian**	1 ♭2 ♭3 ♭4 ♭5 ♭6 𝄫7	1 ♭3 ♭5 𝄫7 = **°7**	dim7	vii°7 into i

Harmonic minor = natural minor with ♮7. The ♭6–7 augmented 2nd is its sound; mode 5 (Phrygian dominant) is why it exists: it gives a minor key its V7.

The chords of harmonic minor

Degree	Chord	Mode	7th chord as intervals	Symbol	In C harmonic minor
1	**i(maj7)**	Harmonic minor	1 ♭3 5 7	**m(maj7)** minor-major 7	Cm(maj7)
2	**iiø7**	Locrian ♮6	1 ♭3 ♭5 ♭7	**m7♭5** half-diminished	Dm7♭5
♭3	**♭IIImaj7♯5**	Ionian ♯5	1 3 ♯5 7	**maj7♯5** major 7 ♯5	E♭maj7♯5
4	**ivm7**	Dorian ♯4	1 ♭3 5 ♭7	**m7** minor 7	Fm7
5	**V7**	Phrygian dominant	1 3 5 ♭7	**7** dominant 7	G7
♭6	**♭VImaj7**	Lydian ♯2	1 3 5 7	**maj7** major 7	A♭maj7
7	**vii°7**	Ultralocrian	1 ♭3 ♭5 𝄫7	**°7** diminished 7	B°7

Symmetric scales

Scale	Formula	Steps	Fits	Versions	Remember
Half-whole diminished	1 ♭2 ♯2 3 ♯4 5 6 ♭7	H W H W H W H W	7♭9, 13♭9 (♭9 ♯9 ♯11 ♮13, natural 5th)	3	Repeats every minor 3rd: four dominants share one scale.
Whole-half diminished	1 2 ♭3 4 ♭5 ♭6 𝄫7 7	W H W H W H W H	°7 (and °(maj7))	3	Same notes as half-whole a semitone lower.
Whole tone	1 2 3 ♯4 ♯5 ♭7	W W W W W W	7♯5, 9♯5 (♮9 ♯11 ♭13, no 5th)	2	Only two exist; no perfect 5th anywhere.
Augmented (hexatonic)	1 ♭3 3 5 ♯5 7	m3 H m3 H m3 H	maj7♯5; mMaj7 colour	4	Two augmented triads a minor 3rd apart (1 3 ♯5 + ♭3 5 7).

Steps: H = half step, W = whole step, m3 = minor 3rd (three half steps).

Other scales worth knowing

Scale	Formula	Steps	Think of it as	Use
Major pentatonic	1 2 3 5 6	W W m3 W m3	Ionian without 4, 7	Major chords, country, melody; over maj7 and dominant
Minor pentatonic	1 ♭3 4 5 ♭7	m3 W W m3 W	Aeolian without 2, ♭6	Minor chords, blues and rock; superimposes widely
Blues scale	1 ♭3 4 ♭5 5 ♭7	m3 W H H m3 W	Minor pentatonic + ♭5	Blues and rock over minor or dominant chords
Major blues scale	1 2 ♭3 3 5 6	W H H m3 W m3	Major pentatonic + ♭3	Country, gospel and blues over a major or dominant I
Bebop dominant	1 2 3 4 5 6 ♭7 7	W W H W W H H H	Mixolydian + 7	Bebop lines over V7: chord tones fall on the beat
Bebop major	1 2 3 4 5 ♯5 6 7	W W H W H H W H	Ionian + ♯5	Bebop lines over maj7 / 6
Bebop Dorian	1 2 ♭3 3 4 5 6 ♭7	W H H H W W H W	Dorian + 3	Bebop lines over m7
Harmonic major	1 2 3 4 5 ♭6 7	W W H W H m3 H	Ionian with ♭6	Major with the borrowed ♭6: the iv sound inside the I chord

5 Superimposition — what and why

What you are doing: playing a shape you already own (a triad or a pentatonic) from a different root, so that its notes land on chosen tensions of the chord underneath. **Why:** it turns "play the 9, ♯11 and 13" into one clear, singable idea that you can find instantly in any key.

The method

1. **Know the chord's scale.** The job decides it: a C7 that goes nowhere (IV7, ♭VII7, subV) is Lydian dominant; a C7 resolving to F is Mixolydian, half-whole or altered depending on how much tension you want.
2. **Pick a triad or pentatonic inside that scale that holds the colour notes you want** — and ideally leaves out the avoid note (the 11 over a major or dominant chord).
3. **Name it by the chord degree it sits on** (II, ♭VI, ♭iii…), not by its letter. "II triad over a dominant" works on every dominant in every key.
4. **Read every note as a tension over the chord root** (D over C7 = 9, ♯11, 13), and end the phrase on a guide tone (3rd or 7th) or let it resolve by half step into the next chord.

Why it works

1. **They are the chord's upper floors.** Stack any chord in thirds past the 7th: 1 3 5 7 9 11 13. A triad built on the 5th, 7th or 9th of the chord is three neighbouring floors of that stack, so it isn't a foreign chord at all: it is the top of a bigger version of the same chord (D over C7 = the top of C13♯11).
2. **They come from the chord's own scale.** Each option is a subset of the scale that fits the chord (Lydian dominant, altered, Dorian…). Pick the structure and you have picked the colour.
3. **They skip the avoid notes.** The note a half step above a chord tone (the 11 over a major or dominant chord) rubs on a long note. Good superimpositions leave it out, or replace it with ♯11.
4. **The ear trusts strong shapes.** A triad or a pentatonic is so familiar that even far-out notes sound deliberate: you are playing a clear idea, not wrong notes.
5. **Tension has somewhere to go.** Over a dominant, altered notes sit a half step from notes of the I chord, so they pull into it when the chord resolves.

Worked example: II major triad over C7

- **The notes.** D F♯ A over C7 = 9, ♯11, 13.
- **Upper floors.** Stack C Lydian dominant in thirds: C E G B♭ D F♯ A (1 3 5 ♭7 9 ♯11 13). The D triad is three neighbouring floors of that stack — the top of the chord, above the 7th: the upper part of the full chord, with the bass and comping supplying the lower notes.
- **From the chord's own scale.** Every note is in C Lydian dominant (C D E F♯ G A B♭), the scale that fits C7, so it adds colour without fighting the chord.
- **Guide tones.** It has neither the 3rd nor the 7th: it is pure colour. Let the bass or comping state the chord, or land on a guide tone at the end of your phrase.
- **Resolving to F.** D steps to C (5); F♯ rises to G (9); A stays (3). Smooth, mostly common-tone motion: a gentle arrival.
- **Sound.** The major triad a step up: 9, ♯11, 13. The Lydian dominant sound used on IV7, ♭VII7 and tritone subs.

Know these cold. Over any dominant: II major triad (9 ♯11 13), VI major triad (13 ♭9 3), ♭VI major triad (♭13 1 ♯9), ♭V major triad (♯11 ♭7 ♭9), ♭iii minor pent. (♯9 ♯11 ♭13 ♭7 ♭9). Over any maj7: II major triad (9 ♯11 13), V major triad (5 7 9). Over any m7: ♭VII major triad (♭7 9 11), IV major triad (11 13 1), ii minor pent. (9 11 5 13 1).

The options, chord by chord

Examples over C; the degree name is what transfers to every key.

Over a major 7 (Xmaj7)

Play	Adds → sounds like	Scale	Why
V major triad G: G B D over Cmaj7	5 7 9 → **maj9**	Ionian	The triad a 5th up adds the 7 and 9 on top of the chord: the plainest, prettiest upper structure. It is three neighbouring floors of the chord's stack of thirds, every note is in the chord's own scale (Ionian), it skips the avoid note.
II major triad D: D F♯ A over Cmaj7	9 ♯11 13 → **maj13♯11**	Lydian	The major triad a step up gives 9, ♯11 and 13: the whole Lydian colour in three notes. It is three neighbouring floors of the chord's stack of thirds, every note is in the chord's own scale (Lydian).
iii minor triad Em: E G B over Cmaj7	3 5 7 → **maj7 (no root)**	Ionian	The minor triad on the 3rd is the chord without its root: 3, 5, 7. Safe, inside, and a smooth way to target the 3rd. It is three neighbouring floors of the chord's stack of thirds, every note is in the chord's own scale (Ionian), it skips the avoid note.
vii minor triad Bm: B D F♯ over Cmaj7	7 9 ♯11 → **maj9♯11**	Lydian	B minor over C: 7, 9, ♯11. Bright and floating. It is three neighbouring floors of the chord's stack of thirds, every note is in the chord's own scale (Lydian).
II major pentatonic D major pentatonic: D E F♯ A B over Cmaj7	9 3 ♯11 13 7 → **maj13♯11**	Lydian	Major pentatonic a step up: 9, 3, ♯11, 13, 7 — no root and no 4th, so nothing clashes. Every note is in the chord's own scale (Lydian).
V major pentatonic G major pentatonic: G A B D E over Cmaj7	5 13 7 9 3 → **maj13**	Ionian	Major pentatonic a 5th up: 5, 13, 7, 9, 3 — Ionian without the avoid note (the 4th). Every note is in the chord's own scale (Ionian), it skips the avoid note.

Over a dominant 7 (X7)

Play	Adds → sounds like	Scale	Why
v minor triad Gm: G B♭ D over C7	5 ♭7 9 → **9**	Mixolydian	The minor triad a 5th up: 5, ♭7, 9. Plain, inside dominant sound. It is three neighbouring floors of the chord's stack of thirds, every note is in the chord's own scale (Mixolydian), it skips the avoid note, its notes pull by half step into the I chord.
II major triad D: D F♯ A over C7	9 ♯11 13 → **13♯11**	Lydian dominant	The major triad a step up: 9, ♯11, 13. The Lydian dominant sound used on IV7, ♭VII7 and tritone subs. It is three neighbouring floors of the chord's stack of thirds, every note is in the chord's own scale (Lydian dominant), its notes pull by half step into the I chord.
VI major triad A: A C♯ E over C7	13 ♭9 3 → **13♭9**	half-whole diminished	The major triad a major 6th up: 13, ♭9, 3. Bright and tense at once; resolves beautifully. Every note is in the chord's own scale (half-whole diminished), its notes pull by half step into the I chord.
♭III major triad E♭: E♭ G B♭ over C7	♯9 5 ♭7 → **7♯9**	half-whole diminished	E♭ over C7: ♯9, 5, ♭7. The bluesy ♯9 sound. Every note is in the chord's own scale (half-whole diminished), its notes pull by half step into the I chord.

Play	Adds → sounds like	Scale	Why
♭V major triad G♭: G♭ B♭ D♭ over C7	♯11 ♭7 ♭9 → **7♭9♯11**	altered / half-whole	G♭ over C7: ♯11, ♭7, ♭9 — the triad of the tritone sub (G♭7), so it bridges C7 and its substitute. It is three neighbouring floors of the chord's stack of thirds, every note is in the chord's own scale (altered / half-whole), its notes pull by half step into the I chord.
♭VI major triad A♭: A♭ C E♭ over C7	♭13 1 ♯9 → **7♯9♭13**	altered	A♭ over C7: ♭13, 1, ♯9. Dark and altered, resolves down to the I. Every note is in the chord's own scale (altered), its notes pull by half step into the I chord.
♭iii minor pentatonic E♭ minor pentatonic: E♭ G♭ A♭ B♭ D♭ over C7	♯9 ♯11 ♭13 ♭7 ♭9 → **7alt**	altered	Minor pentatonic a minor 3rd up: ♯9, ♯11, ♭13, ♭7, ♭9 — five altered notes, no root, no 3rd. Every note is in the chord's own scale (altered), its notes pull by half step into the I chord.
v minor pentatonic G minor pentatonic: G B♭ C D F over C7	5 ♭7 1 9 11 → **9sus4**	Mixolydian	Minor pentatonic a 5th up: 5, ♭7, 1, 9, 11. Suspended, open, great on 7sus4. Every note is in the chord's own scale (Mixolydian), its notes pull by half step into the I chord.

Over a minor 7 (Xm7)

Play	Adds → sounds like	Scale	Why
♭III major triad E♭: E♭ G B♭ over Cm7	♭3 5 ♭7 → **m7 (no root)**	Dorian / Aeolian	E♭ over Cm7: ♭3, 5, ♭7. The chord without its root. It is three neighbouring floors of the chord's stack of thirds, every note is in the chord's own scale (Dorian / Aeolian).
v minor triad Gm: G B♭ D over Cm7	5 ♭7 9 → **m9**	Dorian / Aeolian	The minor triad a 5th up: 5, ♭7, 9. It is three neighbouring floors of the chord's stack of thirds, every note is in the chord's own scale (Dorian / Aeolian).
♭VII major triad B♭: B♭ D F over Cm7	♭7 9 11 → **m11**	Dorian / Aeolian	B♭ over Cm7: ♭7, 9, 11. Open, modern minor. It is three neighbouring floors of the chord's stack of thirds, every note is in the chord's own scale (Dorian / Aeolian).
IV major triad F: F A C over Cm7	11 13 1 → **m13**	Dorian	F over Cm7: 11, 13, 1. The natural 13 is the Dorian note. Every note is in the chord's own scale (Dorian).
i minor pentatonic C minor pentatonic: C E♭ F G B♭ over Cm7	1 ♭3 11 5 ♭7 → **m11**	Dorian / Aeolian	The home minor pentatonic: 1, ♭3, 11, 5, ♭7. Every note is in the chord's own scale (Dorian / Aeolian).
ii minor pentatonic D minor pentatonic: D F G A C over Cm7	9 11 5 13 1 → **m13**	Dorian	Minor pentatonic a step up: 9, 11, 5, 13, 1 — the Dorian pentatonic. Every note is in the chord's own scale (Dorian).
v minor pentatonic G minor pentatonic: G B♭ C D F over Cm7	5 ♭7 1 9 11 → **open m11 (no 3rd)**	Dorian / Aeolian	Minor pentatonic a 5th up: 5, ♭7, 1, 9, 11 — no 3rd, so it floats. Every note is in the chord's own scale (Dorian / Aeolian).

Over a half-diminished (Xm7♭5)

Play	Adds → sounds like	Scale	Why
♭iii minor triad E♭m: E♭ G♭ B♭ over Cm7♭5	♭3 ♭5 ♭7 → **m7♭5 (no root)**	Locrian	E♭m over Cø7: ♭3, ♭5, ♭7. The chord without its root. It is three neighbouring floors of the chord's stack of thirds, every note is in the chord's own scale (Locrian), it skips the avoid note.

Play	Adds → sounds like	Scale	Why
♭VI major triad A♭: A♭ C E♭ over Cm7♭5	♭13 1 ♭3 → **m7♭5♭13**	Locrian	A♭ over Cø7: ♭13, 1, ♭3. Every note is in the chord's own scale (Locrian), it skips the avoid note.
♭VII major triad B♭: B♭ D F over Cm7♭5	♭7 9 11 → **m9♭5**	Locrian ♮2	B♭ over Cø7: ♭7, 9, 11. The natural 9 is the melodic-minor (Locrian ♮2) colour. It is three neighbouring floors of the chord's stack of thirds, every note is in the chord's own scale (Locrian ♮2).
iv minor pentatonic F minor pentatonic: F A♭ B♭ C E♭ over Cm7♭5	11 ♭13 ♭7 1 ♭3 → **m11♭5**	Locrian	Minor pentatonic a 4th up: 11, ♭13, ♭7, 1, ♭3. Every note is in the chord's own scale (Locrian), it skips the avoid note.
♭iii minor pentatonic E♭ minor pentatonic: E♭ G♭ A♭ B♭ D♭ over Cm7♭5	♭3 ♭5 ♭13 ♭7 ♭9 → **m7♭5♭9**	Locrian	Minor pentatonic a minor 3rd up: ♭3, ♭5, ♭13, ♭7, ♭9. Every note is in the chord's own scale (Locrian).

Over a minor-major 7 / minor 6 (Xm(maj7))

Play	Adds → sounds like	Scale	Why
V major triad G: G B D over Cm(maj7)	5 7 9 → **m(maj9)**	melodic minor	G over Cm: 5, 7, 9. The major triad a 5th up is the sound of melodic minor. It is three neighbouring floors of the chord's stack of thirds, every note is in the chord's own scale (melodic minor).
IV major triad F: F A C over Cm(maj7)	11 13 1 → **m6/9**	melodic minor / Dorian	F over Cm: 11, 13, 1. Every note is in the chord's own scale (melodic minor / Dorian).
ii minor pentatonic D minor pentatonic: D F G A C over Cm(maj7)	9 11 5 13 1 → **m6/9**	melodic minor	Minor pentatonic a step up: 9, 11, 5, 13, 1. Every note is in the chord's own scale (melodic minor).

Over a diminished 7 (X°7)

Play	Adds → sounds like	Scale	Why
II major triad D: D F♯ A over C°7	9 ♭5 °7 → **°7 with 9, 11**	whole-half diminished	D over C°7: 9, ♭5, 13(=°7). Any major triad a step above a dim7 note fits. Every note is in the chord's own scale (whole-half diminished).
VII major triad B: B D♯ F♯ over C°7	7 ♭3 ♭5 → **°(maj7)**	whole-half diminished	B over C°7: 7, ♭3, ♭5. The leading-tone triad hidden in the scale. Every note is in the chord's own scale (whole-half diminished).

Daily recall checklist

Be able to reel these off with no instrument and no paper. Tick a line when it is instant.

Circle of fifths

- ☐ Walk the circle both ways from memory (C G D A E B F♯ D♭ A♭ E♭ B♭ F) with the relative minor inside each.
- ☐ The sharps or flats of any key, in order (F C G D A E B / B E A D G C F).
- ☐ Name the key from any signature: last sharp + a semitone; the second-to-last flat.

Modal interchange in a major key

- ☐ The core borrowed chords, each with its source mode(s): ivm7, ♭VImaj7, ♭VII7, ♭IIImaj7, iiø7, vm7, ♭IImaj7, II7, ♭VIImaj7, im7.
- ☐ What each parallel mode lends — e.g. Aeolian: im7, iiø7, ♭IIImaj7, ivm7, vm7, ♭VImaj7, ♭VII7.
- ☐ The job of each core chord: minor plagal, deceptive ♭VI, backdoor ♭VII7, Mario cadence, Neapolitan ♭II, Lydian II.
- ☐ The complete set: run down the list and check you know every one and where it comes from.
- ☐ One stock move a day, played in today's key.

Modal interchange in a minor key

- ☐ The core borrowed chords with sources: V7, vii°7, Imaj7, IV7, iim7, ♭IImaj7, i(maj7), viø7, I7, ♭VI7.
- ☐ Which come from the parallel major (Picardy I), from Dorian (IV7), from harmonic minor (V7, vii°7) and from melodic minor (iim7, IV7, V7, i(maj7)).

Interval structure of every scale

- ☐ The formula of every major mode, by the brightness ladder.
- ☐ The formula of every melodic-minor and harmonic-minor mode, and its 7th chord as intervals (1 ♭3 ♭5 ♭7 = m7♭5 …).
- ☐ The seven chords of melodic minor and of harmonic minor, as numerals and as interval structures.
- ☐ Half-whole, whole-half, whole-tone and augmented: the step pattern and what each fits.
- ☐ Major and minor pentatonic, blues, major blues and the bebop scales as "a scale you know, plus or minus a note".

Superimposition

- ☐ The four steps: chord scale → shape that holds the colour → name it by degree → read it as tensions.
- ☐ The why: upper floors of the chord, inside the chord's scale, skips the avoid note, strong shapes, tension that resolves.
- ☐ Over any dominant: II, VI, ♭VI, ♭V triads and the ♭iii minor pentatonic — the tensions and the sound of each.
- ☐ Over any maj7: II and V triads; over any m7: ♭VII and IV triads and the ii minor pentatonic.

> **The daily routine (10 minutes):** pick today's key by walking the circle. Say its signature. Name the core borrowed chords of the major and the minor key on that tonic, with sources. Recite one scale family's formulas and chords. Then name the superimpositions over its V7. Anything slower than a second goes on tomorrow's list.